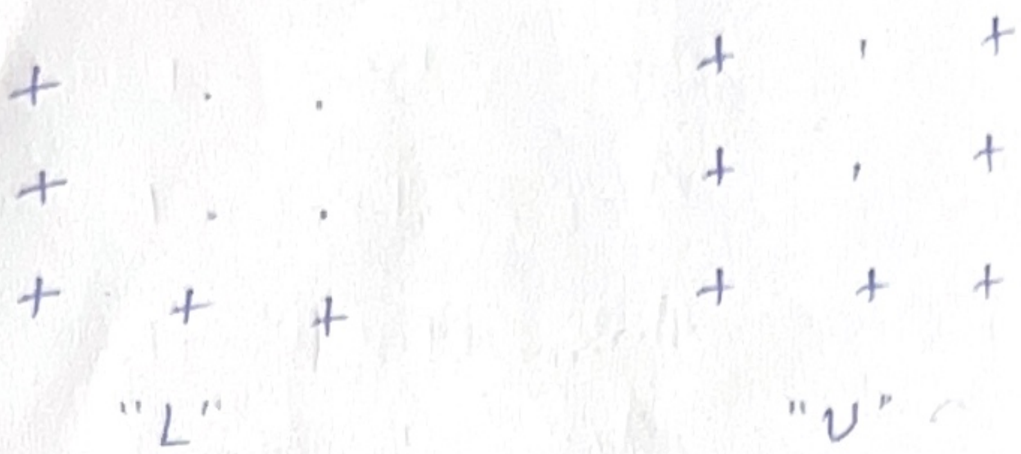
Find the weights required to perform following classifications of given input patterns using Hebb rule. The input are "1" then "+" symbol is present and "-1" then "." is present. The "L" pattern belongs to the class (target value +1) and the "U" pattern doesn't belong to the class (target value -1).



Patterns	Inputs										Target Y
	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	b	
L	1	-1	-1	1	-1	-1	1	1	1	1	1
U	1	-1	1	1	-1	1	1	1	1	1	-1

Initially $w_1 = w_2 = w_3 = w_4 = w_5 = w_6 = w_7 = w_8 = w_9 = b = 0$

Case 1

$w_i(new) = w_i(old) + x_i y$

$w_1(new) = 0 + (1 \times 1) = 1$

$w_2(new) = 0 + (-1 \times 1) = -1$

$w_3(new) = 0 + (-1 \times 1) = -1$

$w_4(new) = 0 + (1 \times 1) = 1$

$w_5(new) = 0 + (-1 \times 1) = -1$

$w_6(new) = 0 + (-1 \times 1) = -1$

$w_7(new) = 0 + (1 \times 1) = 1$

$w_8(new) = 0 + (1 \times 1) = 1$

$w_9(new) = 0 + (1 \times 1) = 1$

$b(new) = b(old) + y = 0 + 1 = 1$

$\Delta w_i = x_i y$

$\Delta w_1 = 1 \times 1 = 1$

$\Delta w_2 = -1 \times 1 = -1$

$\Delta w_3 = -1 \times 1 = -1$

$\Delta w_4 = 1 \times 1 = 1$

$\Delta w_5 = -1 \times 1 = -1$

$\Delta w_6 = -1 \times 1 = -1$

$\Delta w_7 = 1 \times 1 = 1$

$\Delta w_8 = 1 \times 1 = 1$

$\Delta w_9 = 1 \times 1 = 1$

$\Delta b = y = 1$

$[w_1\ w_2\ w_3\ w_4\ w_5\ w_6\ w_7\ w_8\ w_9\ b] = [1\ -1\ -1\ 1\ -1\ -1\ 1\ 1\ 1\ 1]$

Case II

$w_1(new) = 1 + (1 \times -1) = 0$

$w_2(new) = -1 + (-1 \times -1) = 0$

$w_3(new) = -1 + (1 \times -1) = -2$

$w_4(new) = 1 + (1 \times -1) = 0$

$w_5(new) = -1 + (-1 \times -1) = 0$

$w_6(new) = -1 + (1 \times -1) = -2$

$w_7(new) = 1 + (1 \times -1) = 0$

$w_8(new) = 1 + (1 \times -1) = 0$

$w_9(new) = 1 + (1 \times -1) = 0$

$\Delta w_1 = 1 \times -1 = -1$

$\Delta w_2 = -1 \times -1 = 1$

$\Delta w_3 = 1 \times -1 = -1$

$\Delta w_4 = 1 \times -1 = -1$

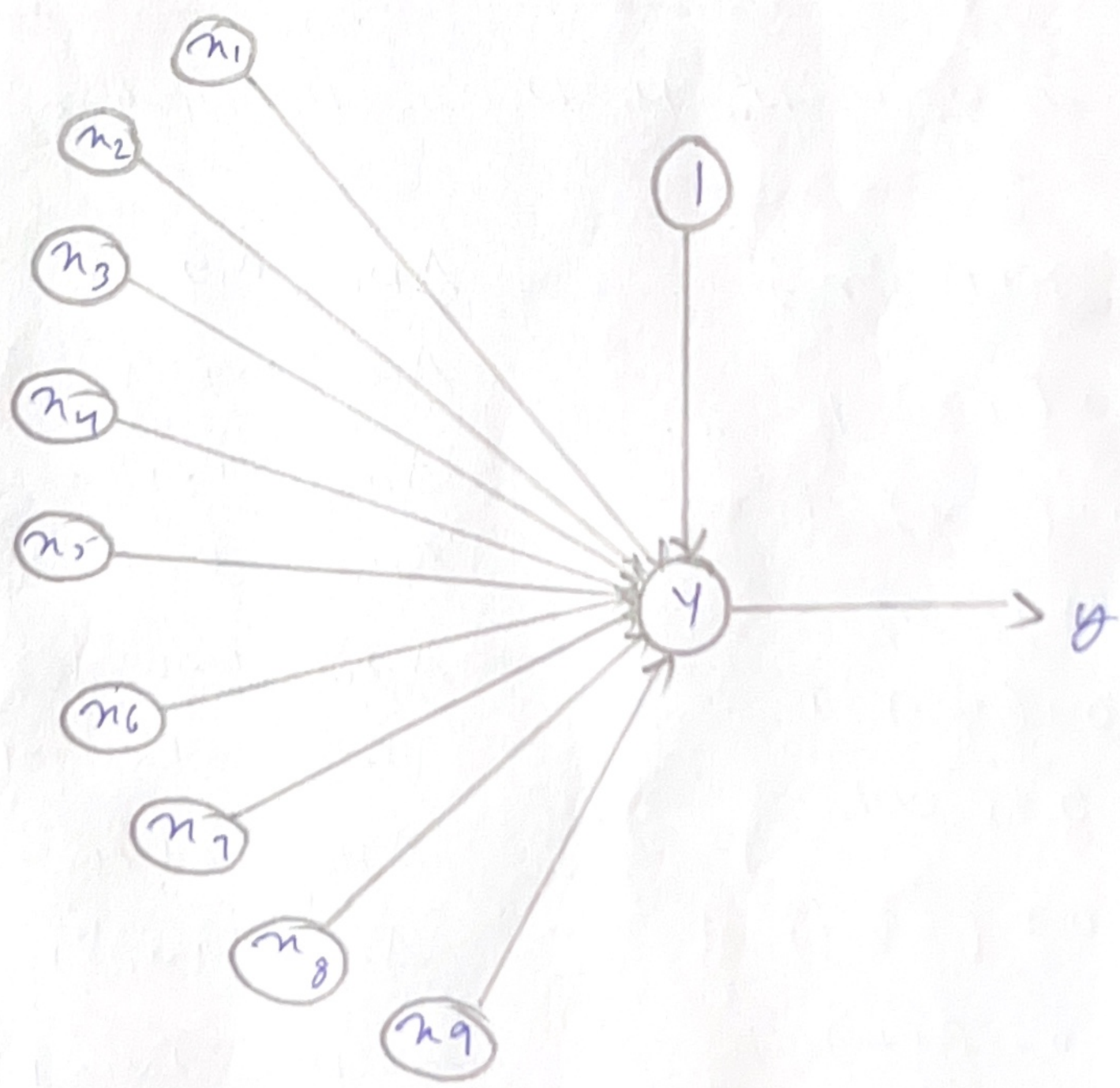
$\Delta w_5 = -1 \times -1 = 1$

$\Delta w_6 = -1 \times -1 = 1$

$\Delta w_7 = 1 \times -1 = -1$

$\Delta w_8 = 1 \times -1 = -1$

$\Delta w_9 = 1 \times -1 = -1$

$\Delta b = y = -1$

$[w_1\ w_2\ w_3\ w_4\ w_5\ w_6\ w_7\ w_8\ w_9\ b] = [0\ 0\ -2\ 0\ 0\ -2\ 0\ 0\ 0\ 0]$